
Welcome & Module Induction

What we learned previously: In Module 4, we learned how to track and protect our code using Git, allowing us to collaborate safely and revert mistakes using terminal commands.

What we are learning today: We are going to write code that controls the Operating System itself. We will learn how to programmatically move, rename, and analyze thousands of files at once using Python's built-in `os`, `shutil`, and `pathlib` modules.

Why this will be helpful: In AI and Computer Vision, you rarely work with just one file. You work with datasets containing 50,000 thermal images. If you need to separate the images into "Training" and "Validation" folders, or delete every corrupted image under 5KB, you cannot do it by hand. Today, you will learn how to write a script that does it in milliseconds.

The os and shutil Modules (The Heavy Lifters)

The os Module: Used for interacting with the operating system's structure. You use it to create directories, list files, and delete empty folders.

The shutil Module: "Shell Utilities." This is the heavy lifter for data movement. You use it to copy files, move files across drives, or delete entire directories that are full of data (rmtree).

pathlib (The Modern Standard)

The Problem: Windows uses backslashes for paths (C:\data\images), while Mac and Linux (and most servers) use forward slashes (/data/images). If you hardcode slashes, your script will crash when someone else runs it.

The Solution: pathlib. Introduced in modern Python, it treats paths not as "text strings", but as smart *objects*. It automatically uses the correct slash for whatever operating system it is currently running on.

File Permissions & Metadata

Beyond the Data: A file is more than just the image or text inside it. The Operating System stores "Metadata" (data about the data).

What we can read: With Python, we can instantly read a file's exact size in bytes, the exact second it was created or modified, and whether it is "Read-Only." This is critical for data validation pipelines.